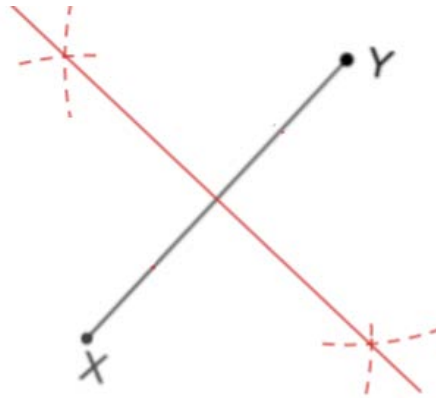
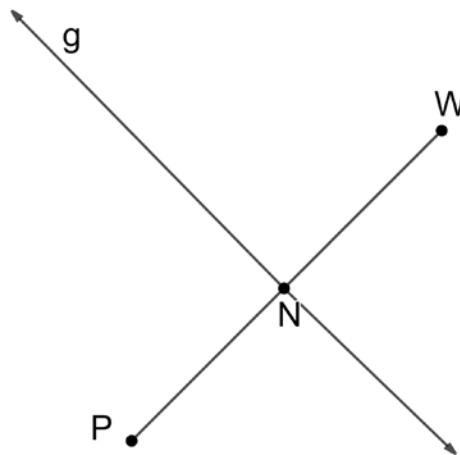


1. Use a compass and a straightedge to construct a perpendicular bisector of $\overline{XY}$.



Line g is a perpendicular bisector of $\overline{PW}$ at point N .



2. If $PN = 4x + 17$ and $WN = 9x - 20$, find the value of x .

$$4x + 17 = 9x - 20$$

$$37 = 5x$$

$$x = \frac{37}{5} = 7.4$$

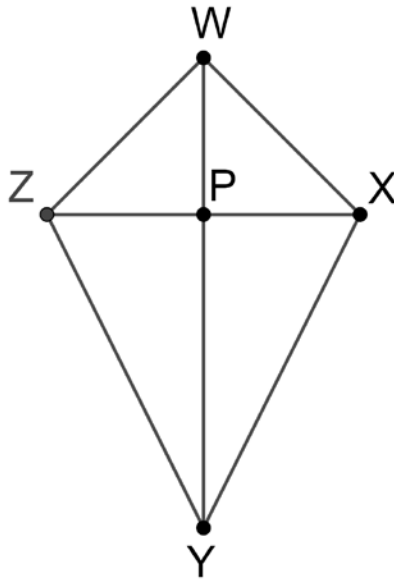
3. How long is $\overline{PN}$?

$$\overline{PN} = 4x + 17 = 4(7.4) + 17 = 46.6$$

4. How long is $\overline{PW}$?

$$\overline{PW} = \overline{PN} + \overline{WN} = 4x + 17 + 9x - 20 = 13x - 3 = 13(7.4) - 3 = 93.2$$

A quadrilateral is shown where $\overline{WY}$ is a perpendicular bisector of $\overline{ZX}$.



5. What angle is congruent to $\angle WZP$?
 $\angle WXP$
6. What angle is congruent to $\angle ZWP$?
 $\angle XWP$
7. Which angle(s) are congruent to $\angle WPZ$?
 $\angle WPX, \angle XPY, \angle ZPY$
8. What kind of triangle is $\triangle XYZ$?
Isosceles

$\overline{SL}$ is a perpendicular bisector of $\overline{RT}$, intersecting at point N . The length of $\overline{SR}$ is $\frac{2}{3}x - 40$ and the length of $\overline{ST}$ is $\frac{1}{4}x + 12$.

9. Find the value of x .

$$\frac{2}{3}x - 40 = \frac{1}{4}x + 12$$

$$\frac{5}{12}x = 52$$

$$x = \frac{624}{5} = 124.8$$
10. Find the length of $\overline{ST}$.

$$\overline{ST} = \frac{1}{4}x + 12 = \frac{1}{4}(124.8) + 12 = 31.2 + 12 = 43.2$$